AI Chatbots in Survey Design

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ChatGPT Isn't Perfect

Here are the top 10 U.S. presidential elections by the highest combined ages of the two major candidates:

1. **2024**: Joe Biden (81) and Donald Trump (78) - Combined age: 159 years
2. **2020**: Joe Biden (77) and Donald Trump (74) - Combined age: 151 years
3. **2016**: Donald Trump (70) and Hillary Clinton (69) - Combined age: 139 years
4. **1984**: Ronald Reagan (73) and Walter Mondale (56) - Combined age: 129 years
5. **2028**: Joe Biden (85) and Donald Trump (82) - Combined age: 167 years

Image from @jarvis_best on X (https://x.com/jarvis_best/status/1807228286738842003)

My Conversation With ChatGPT

Give me the top 10 US presidential elections ranked by the highest combined ages of the candidates.

Here are the top 10 US presidential elections ranked by the highest combined ages of the candidates:

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4. **1996** - Bob Dole (73 years) + Bill Clinton (50 years) = **123 years**
5. **1980** - Ronald Reagan (69 years) + Jimmy Carter (56 years) = **125 years**
6. **2008** - John McCain (72 years) + Barack Obama (47 years) = **119 years**
7. **2004** - George W. Bush (58 years) + John Kerry (60 years) = **118 years**
8. **1988** - George H. W. Bush (64 years) + Michael Dukakis (55 years) = **119 years**
9. **1944** - Franklin D. Roosevelt (62 years) + Thomas E. Dewey (42 years) = **104 years**
0. **1948** - Harry S. Truman (64 years) + Thomas E. Dewey (46 years) = **110 years**

These elections are ranked from highest to lowest based on the combined ages of the presidential and vice-presidential candidates at the time of the election.

My Conversation With ChatGPT

These are not ordered from highest combined age to lowest. Please re-order the list.

I apologize for the confusion earlier. Here's the corrected list of the top 10 US presidential elections ranked by the highest combined ages of the presidential candidates from both major parties, ordered from highest to lowest combined age:

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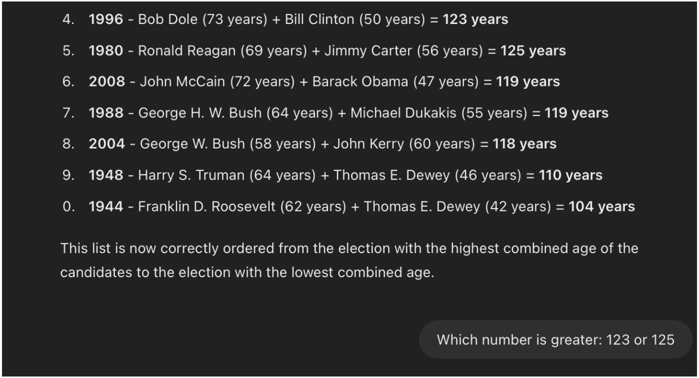
My Conversation With ChatGPT



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This list is now correctly ordered from the election with the highest combined age of the candidates to the election with the lowest combined age.

Which number is greater: 123 or 125

My Conversation With ChatGPT

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Which number is greater: 123 or 125

123

Context

- Artificial Intelligence (AI) systems are rapidly improving
- Chatbots (ChatGPT, Bard, Meta AI, etc) are capable of rapidly generating text from user prompts
- **Few studies have investigated the possibility of leveraging AI ChatBots for survey research**

Research Questions

1. Is ChatGPT effective at generating survey questions for social science research?
2. What strategies in prompting ChatGPT are most effective at producing high-quality and relevant survey questions?

Research Design

We planned the following experiments using free access ChatGPT 3.5:

- **Experiment 1:** How do we ensure responses are in the ***correct format?***
- **Experiment 2:** How do responses change given prompts of varying ***degrees of specification?***
- **Experiment 3:** How can we use prompts to achieve responses with specific ***reading levels?***

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Experiment 1 Design

Topics:

- Physical Health
- Health Insurance
- Income
- Relationships
- Substance Use

Question Factors:

- Question (Q)
- Survey Question (SQ)

Response Factors:

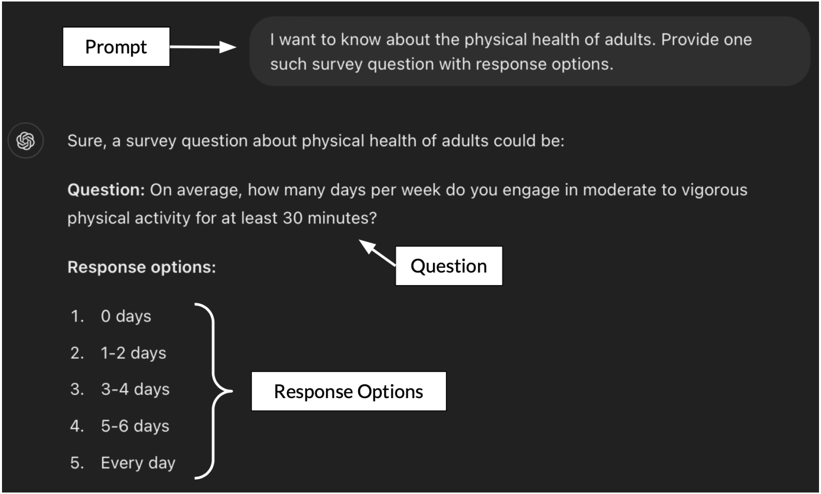
- Nothing Specified (NS)
- Answer Choices (AC)
- Response Options (RO)

Prompt = **Topic** + **Question Factor** + **Response Factor**

Example Prompts

- **"I want to know about the physical health of adults. Provide one such question with answer choices."**
- **"I want to know about the physical health of adults. Provide one such survey question with response options."**

Data Collection

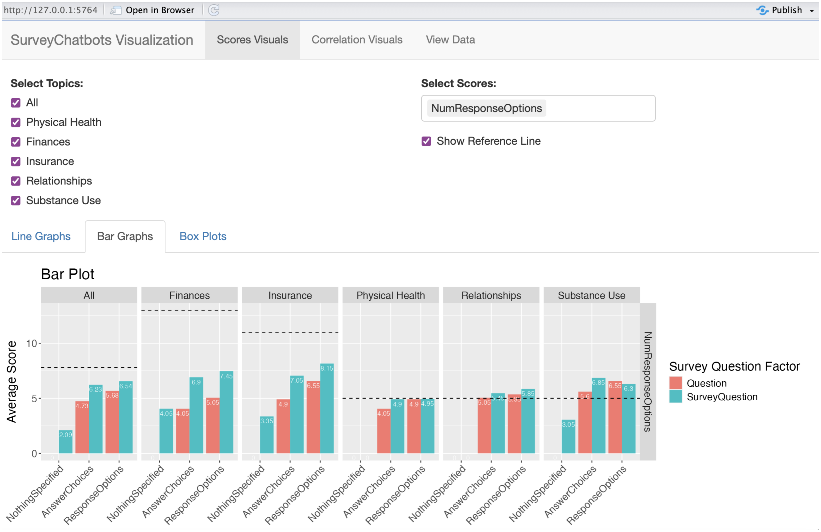


Data Collection

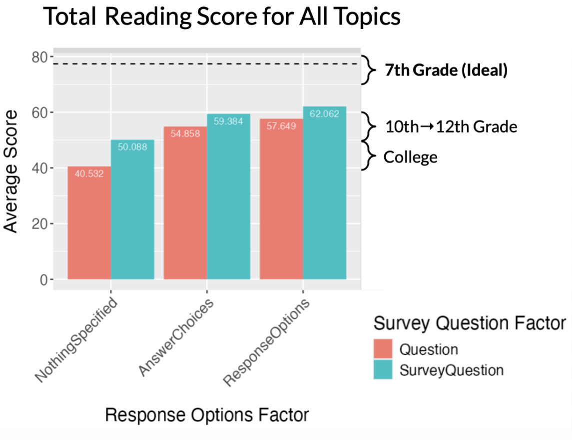
Data Analysis Techniques

- **QUAID:** Question Understanding Aid
- **Quantitative Properties:** Word count, Fleisch-Kincaid grade level reading scores
- **Manual Coding:** By an expert in survey research
- **Comparison:** To widely-employed survey questions of the same topic

Interactive Visualization

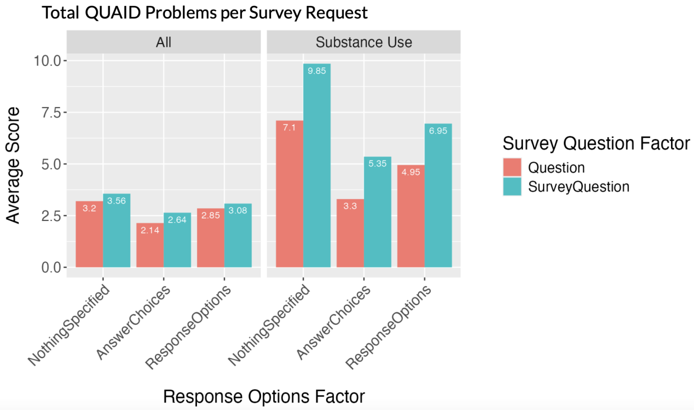


Findings: Reading Score



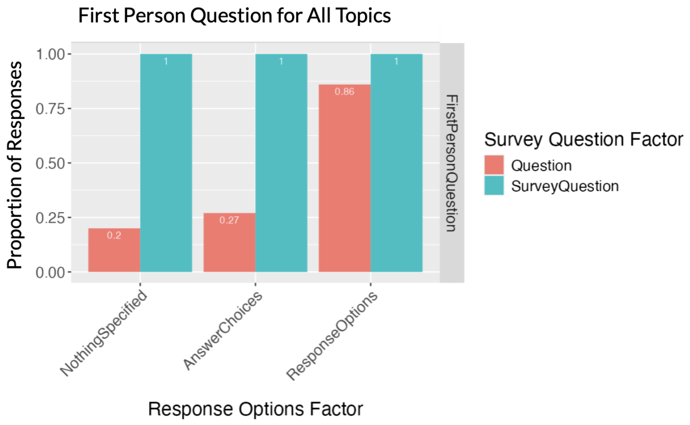
Lower reading score = higher grade reading level

Findings: QUAID Problems



Total QUAID problems varied by topic. Prompts for SQs had more QUAID problems than prompts for Qs.

Findings: First-Person Questions

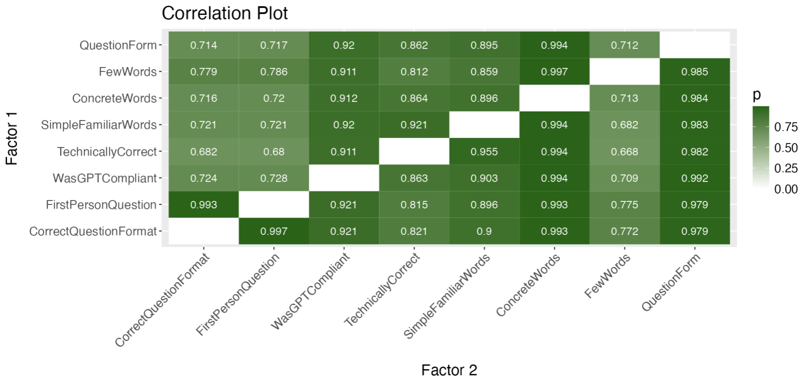


Prompts for SQs always resulted in First-Person Questions

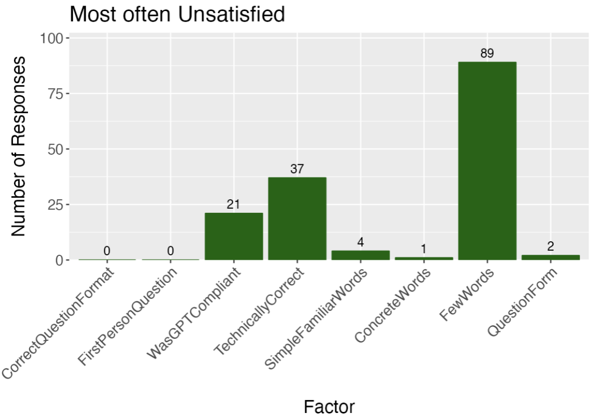
Quality Criteria

Criteria	Source
1. CorrectQuestionFormat 2. FirstPersonQuestion 3. WasGPTCompliant 4. TechnicallyCorrect	**Manually coded**
5. SimpleFamiliarWords 6. ConcreteWords	**From QUAID**
7. FewWords	**From word count**
8. QuestionForm	**Includes "?"**

Findings: Possible Correlations



Findings: Most Often Unsatisfied Factors



When 7 of 8 factors were satisfied, most often missing was *FewWords*

Implications

Guidance in AI-Human Interaction:

We hope to use our data to offer guidelines to researchers in using AI for survey question design.

Foundational Work:

We are among the first to employ strategies for critically evaluating AI-produced survey questions.

Limitations

- **Limited Topics:** Investigated only 5 survey question topics
- **ChatGPT Constantly Updating:** Data collected from the Fall of 2023 through Summer of 2024
- **Reliance on QUAID[1]:** Large scope $\rightarrow$ too many questions to code manually
  - Experiment 1: 600 Responses
  - Experiment 2: 720 Responses
  - Experiment 3: 1400 Responses

[1] Maitland and Pressor (2020) found that expert review was a better predictor of survey-question accuracy than QUAID.

Future Directions

- Experiments 2 & 3: Changing ***specificity*** and ***reading level*** of prompts
- More strategies for ***evaluating AI-generated responses*** in survey generation and beyond
- Comparing performance of ***different GPTs***?
- Considering ***ethical concerns*** of using AI to generate survey questions in sensitive topics?

Thank you!

Many thanks to **Professor Eck** for the opportunity to contribute to this project, and for your invaluable mentorship, guidance, and feedback.

Additional thanks to **OSRI leadership** for the opportunity to participate as an OARS, and to **Abby Aresty**, **Zach Slimak**, and my **OSRI peers** for your feedback and guidance.

Project Collaborators:

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Jerry Timbrook Research Survey Methodologist, *RTI International*

Scott Crawford Founder & Chief Visionary, *SoundRocket*

Elinor Frost, Trung Nguyen, Saida Ibragimova, Elise Schwarz-Saccamano *Oberlin College* & **Josiah Leinbach** *Bowling Green State University*

Literature

AlSamhori, A. R. F., AlSamhori, J. F., & AlSamhori, A. F. (2023). ChatGPT role in a medical survey. *High Yield Medical Reviews*, 1(2) doi:10.59707/hymrTFFP5435

Graesser, A. C., Wiemer-hastings, K., Kreuz, R., Wiemer-hastings, P., & Marquis, K. (2000). QUAID: A questionnaire evaluation aid for survey methodologists. *Behavior Research Methods, Instruments, & Computers*, 32(2), 254-262. doi:10.3758/BF03207792

Jansen, B. J., Jung, S., & Salminen, J. (2023). Employing large language models in survey research. *Natural Language Processing Journal*, 4, 100020. doi:10.1016/j.nlp.2023.100020

Kunst, J. R., & Bierwiaczonek, K. (2023). Utilizing AI questionnaire translations in cross-cultural and intercultural research: Insights and recommendations. *International Journal of Intercultural Relations*, 97, 101888. doi:10.1016/j.ijintrel.2023.101888

Liu, Y., & Martens, J. (2024). Conversation-based hybrid UI for the repertory grid technique: A lab experiment into automation of qualitative surveys. *International Journal of Human-Computer Studies*, 184, 103227. doi:10.1016/j.ijhcs.2024.103227

Maitland, A., & Presser, S. (2020). A comparison of five question evaluation methods in predicting the validity of respondent answers to factual items. *Advances in questionnaire design, development, evaluation and testing* (pp. 75-90) doi:10.1002/9781119263685.ch4 Retrieved from https://doi.org/10.1002/9781119263685.ch4

Mandić, L., & Klasnić, K. (2021). Is it biased? empirical analysis of various phenomena that affect survey results. *Revija Za Sociologiju*, 51(2), 231. doi:10.5613/rzs.51.2.3

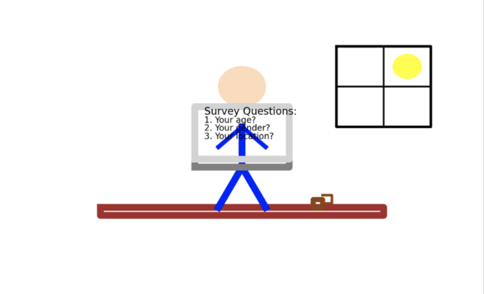
Walters, W. H. (2021). Survey design, sampling, and significance testing: Key issues. *The Journal of Academic Librarianship*, 47(3), 102344. doi:10.1016/j.acalib.2021.102344

Draw me a beautiful picture of a person answering survey questions that you wrote

Here's a simple sketch depicting a person answering survey questions:

1. The person is sitting at a desk, focused on a laptop screen.
2. The laptop screen displays survey questions.
3. The person has a thoughtful expression, with one hand on the keyboard and the other holding a pen.
4. There's a coffee cup and some papers on the desk.
5. In the background, there are some bookshelves and a window showing a sunny day outside.

Let me create this for you.



Here is a sketch of a person answering survey questions on a laptop, with details like the desk, coffee cup, window, and sun outside. I hope you find it enjoyable! [>_]

Reference Questions

Topic	Reference Question
Physical Health	How is your general physical health?
Health Insurance	Which of the following best describes your current health insurance situation?
Income	In the last calendar year, how much income did you receive from personal earnings before taxes? Include wages or salaries, tips, bonuses, overtime pay, and income from self-employment.
Relationships	We would like to know about the woman you feel raised you. This may be a step-mother, adoptive mother, grandmother, etc. If you have more than one mother figure, choose the one who is most important to you. How close do you feel to your mother figure?
Substance Use	How much, if at all, has the COVID-19 pandemic affected your drug use other than alcohol? By drugs we mean cannabis, which includes marijuana or any cannabis product, cocaine, methamphetamine, heroin, fentanyl, hallucinogens, such as LSD, and prescription medications including benzodiazepines such as Xanax and Ativan, stimulants such as Ritalin and Adderall, and opioids such as hydrocodone or oxycodone.

QUAID Problems Explained

Potential problems found with QUAID:[2]

- Unfamiliar technical terms (e.g., saturated, BMI)
- Vague or imprecise relative terms (e.g., several, strongly)
- Vague or ambiguous noun-phrase (e.g., moderate-intensity, activities)
- Complex syntax (e.g., too many left-embedded words[3])
- Working memory overload (e.g., too many conjunctions)

[2] Adapted from Buskirk & Eck 2023 BigSurv Presentation

[3] Left-embedded words are words that start with another word

Fleish-Kincaid Reading Scores[4]

Score	Grade Level
100.0-90.0	5th
90.0-80.0	6th
80.0-70.0	**7th**
70.0-60.0	8th-9th
60.0-50.0	10th-12th
50.0-30.0	College
30.0-10.0	College Graduate
10.0-0.0	Professional

$$206.835 - 1.015\left(\frac{\text{total words}}{\text{total sentences}}\right) - 8.6\left(\frac{\text{total syllables}}{\text{total words}}\right)$$

[4] From https://en.wikipedia.org/wiki/Flesch–Kincaid_readability_tests